

## Problem 10.22

If

$$a(t) = 1 + 0.01(t^2 + t),$$

calculate  $\delta_5$ .

## Solution

The force of interest at time  $t$  is

$$\delta_t = \frac{a'(t)}{a(t)}.$$

Differentiating the accumulation function gives

$$a'(t) = 0.01(2t + 1) = 0.02t + 0.01.$$

At time  $t = 5$ ,

$$a'(5) = 0.02(5) + 0.01 = 0.11$$

and

$$a(5) = 1 + 0.01(5^2 + 5) = 1 + 0.01(30) = 1.30.$$

Therefore,

$$\begin{aligned}\delta_5 &= \frac{a'(5)}{a(5)} \\ &= \frac{0.11}{1.30} \\ &\approx 0.08461538462.\end{aligned}$$

Thus,

$$\boxed{\delta_5 \approx 8.4615\%}.$$